COM3031-Probabilistic Machine Learning Workshop: Information Theory and Hidden Markov Models

Overview

This workshop covers two topics: basic concepts in information theory and hidden Markov models (HMMs). In the first part, you will explore self-information, Shannon entropy, and the relationship between probability and uncertainty. In the second part, you will estimate HMM parameters and infer hidden state sequences.

1 Information Theory

Instructions

1. Calculate the self-information for a fair coin flip and a fair die roll using
$$I(x) = -\log_2(p).$$
2. Plot the relationship between probability and self-information for probability values ranging from 0.1 to 1.0.
3. Compute the entropy of a fair six-sided die using Shannon's entropy formula
$$H(X) = -\sum p(x)\log_2 p(x),$$
and verify your result using the `scipy.stats.entropy()` function.
4. Compute the entropy for binary probability distributions of the form $[p, 1-p]$, where p varies from 0 to 0.5, and plot entropy as a function of probability.

2 Hidden Markov Models

The problem description is given as follows: assuming that you have a document with the vocabulary of English words, the aim is to find the hidden sequence of two hidden states: consonants and vowels. A document written in the English language has been provided to you on ELE. The tasks are given as follows.

Instructions

1. Make a list of all the alphabets from the document. In the document, there are 1743 alphabets.

2. Assume there are two hidden states: 0 (for consonants) and 1 (for vowels). Use the Baum-Welch algorithm to learn the parameters, i.e., the initial probability vector, the transition probability matrix and the emission probability matrix of the hidden Markov model.

3. From the emission probability matrix, find if an alphabet is a vowel or a consonant.

4. Given the HMM and its parameters, use the Viterbi algorithm to find the sequence of two hidden states.

5. Compute the probability of the given observation sequence using the Forward Algorithm.

For demonstration of the first two tasks, I will be using the provided Jupyter notebook and the `mchmm` library: `https://pypi.org/project/mchmm/`. Please check the instructions on the web link to install the library. You will be working on the last two tasks. I will publish the solution in Python by this week.